

Learn How to Visualise Graph Theory

The focus now shifts from Maxima's computation features to its capabilities in visualising graphs. In this article, the author discusses the drawing of graphs in Maxima.



This 23rd article on the mathematical journey through open source introduces graph theory with visuals, using the graphs package of Maxima.

Graphs here refer to the structures formed using some points (or vertices) and some lines (or edges) connecting them. A simple example would be a graph with two vertices, say '0' and '1', and an edge between '0' and '1'. If all the edges of a graph have a sense of direction from one vertex to another, typically represented by an arrow at their end(s), we call that a directed graph with directed edges. In such a case, we consider the edges as not between two vertices but from one vertex to another. Directed edges are also referred to as arcs. In the above example, we could have two directed arcs, one from '0' to '1', and another from '1' to '0'. Figures 1 and 2 show an undirected and a directed graph, respectively.

Graph creation and visualisation

Now, how did we get those figures? We got them by using the graphs package of Maxima, which is loaded by invoking `load(graphs)` at the Maxima prompt. In the package, vertices are represented by whole numbers 0, 1, ... and

an edge is represented as a list of its two vertices. Using these notations, we can create graphs using the `create_graph(<vertex_list>, <edge_list>[, <directed>])` function. And then `draw_graph(<graph>[, <option1>, <option2>, ...])` will draw the graph pictorially. The code snippet below shows this in action:

```
$ maxima -q
(%i1) load(graphs)$
...
0 errors, 0 warnings
(%i2) g: create_graph([0, 1], [[0, 1]])$
(%i3) dg: create_graph([0, 1], [[0, 1]], directed=true)$
(%i4) draw_graph(g, show_id=true, vertex_size=5, vertex_color=yellow)$
(%i5) draw_graph(dg, show_id=true, vertex_size=5, vertex_color=yellow)$
```

The '`show_id=true`' option draws the vertex numbers, and `vertex_size` and `vertex_color` draw the vertices with the corresponding size and colour.